

EDITOR'S PAGE



Beyond Information Asymmetry in Medicine

The Changing Nature of Medical Authority and Expertise



Harlan M. Krumholz, MD, SM, *Editor-in-Chief, JACC*; Harold H. Hines, Jr Professor of Medicine, Yale School of Medicine, New Haven, Connecticut, USA

When my father graduated from medical school in 1960 and began seeing patients, they came with questions that they had few other ways to answer. They knew their bodies, lives, and goals, but medical knowledge was concentrated within the profession. During my career, the internet began to change that dynamic. Now, my daughter is an intern, entering a world in which patients can obtain not only information, but explanations, interpretations, and recommendations before they meet their clinician.

In a 2026 national survey, 32% of U.S. adults reported using artificial intelligence (AI) for health information or advice.¹ Almost 1 in 5 had used it to explain tests, laboratory results, or diagnoses, and an equal share had used it to compare treatment options. One in 6 had used it to help decide whether to seek care. Patients are no longer merely searching. They are receiving answers before entering the examination room.

Information asymmetry in medicine has never been only about who knows more facts. It also concerns whose claims are treated as authoritative. Conversation analysis has shown that physicians and patients actively negotiate those boundaries.² For much of modern medicine, privileged access to knowledge reinforced the physician's expertise and authority. That basis of authority is receding.

Search engines gave patients documents. Large language models give them an interlocutor. They can interpret information and respond to follow-up questions. Patients may arrive not simply with information, but with an interpretation and a proposed plan.

Physicians, too, are using generative and evidence-grounded clinical assistants. Patients and clinicians may increasingly enter the room having consulted systems built on overlapping evidence and related technology. The informational foundations of the encounter are converging.

Sometimes the profession mounts a defense by cataloging things machines cannot yet do, but that is a fragile strategy. Future systems will know the longitudinal record, elicit the patient's priorities, estimate uncertainty, and recommend the course most consistent with those priorities. In bounded diagnostic tasks, some systems may already outperform physicians.³ What, then, will physicians contribute when sophisticated synthesis, contextualization, and perhaps even judgment are increasingly available through AI?

The durable distinction is not between what a machine can know and what a clinician can know. It is between an answer and a relationship.

The patient-clinician relationship is grounded in fidelity, a sustained and accountable commitment to place the patient's welfare, values, and interests at the center of care; deploy every available tool on that person's behalf; and remain present as circumstances change. Fidelity means being honest with patients, respecting their choices, and remaining committed to them when the path becomes difficult.

An AI system may eventually perform many functions associated with clinical expertise. The profession's distinctive claim cannot be a permanent monopoly on those capabilities. A model may generate an excellent recommendation. A health professional enters a relationship of obligation and

becomes answerable for how it is used. The future of expertise may be defined less by what knowledge physicians possess than by the fidelity with which they serve.

This shift has profound implications for medicine's place in society. Credentials may establish a presumption of competence, but they no longer convey sole authority. Patients can check our claims, compare our explanations, record or transcribe our conversations, and return with another interpretation. Our reasoning is becoming more visible and testable.

"The future of expertise may be defined less by what knowledge physicians possess than by the fidelity with which they serve."

We may be tempted to look back wistfully to a time when the physician stood as the principal source and interpreter of medical knowledge, an oracle whose answers were rarely questioned. But nostalgia is not a strategy for the future, and some aspects of that authority deserved to be challenged. Information asymmetry could protect weak reasoning, discourage questions, and permit position to substitute for explanation. We should not seek to recover unquestioned authority. We should seek to deserve informed trust.

Doing so will require more of physicians, not less. Fluency can masquerade as wisdom. Whatever the source of an answer, clinical expertise often begins by asking what is missing and where uncertainty lies. Deep knowledge remains essential because the more fluent the machines become, the more physicians must know when to disagree with them.

That need is already evident. In a randomized vignette study involving 457 clinicians, standard AI modestly improved diagnostic accuracy, but systematically biased AI reduced it by 11.3 percentage points. Providing image-based explanations did not meaningfully mitigate the harm.⁴ An accompanying editorial described automation bias, the tendency to defer to automated recommendations even when contradictory or clinically nonsensical information should raise concern.⁵ Simply placing a clinician "in the loop" is not enough.

Clinicians will need to question model outputs and know when their own judgment should prevail. This is not entirely new. Physicians already weigh advice

from colleagues, tests, and guidelines rather than simply accept it. AI makes that discipline more important. We must also guard against the erosion of, or failure to develop, the skills needed to exercise independent judgment.

The encounter must evolve as well. We will need to listen without becoming defensive or resentful, make our reasoning visible, communicate uncertainty, and recognize when a patient or a model has identified something we missed. A useful opening question may be, "What have you learned so far, and what do you think it means?" Effective communication begins by uncovering the patient's understanding and values, providing information clearly, establishing credibility, and confirming shared understanding.⁶

Medical education must prepare physicians to work critically with AI while preserving the scientific foundations of judgment. Training should teach clinicians to use these tools critically, and know when to resist them.⁵ It should also strengthen the human qualities on which good care depends, including humility, communication, and fidelity.⁷ These are not soft skills added after the science. They are how science becomes care.

There is an optimistic future within reach. AI can help patients understand their conditions and arrive better prepared to participate in their care. In the KFF survey, 58% of adults who used AI for physical health information or advice later followed up with a clinician.¹ A recent qualitative study likewise found that participants wanted AI to support clinicians rather than replace them, with attention to relational engagement, structural support, reliable performance, and clinician responsibility.⁸

Better-informed patients are not a threat to medical authority. They are an opportunity to create the fuller partnership medicine has long promised. The profession should welcome their preparation, engage seriously with the questions they bring, and place the best available intelligence in the service of each person seeking our care.

My father practiced in a world in which physicians were the principal gateway to medical knowledge. My daughter is entering one in which patients and clinicians may consult the same class of systems before they meet. In that future, patients can be better informed, better prepared, and more capable of participating fully in their care.

It will also ask more of the profession. We will need to earn trust through competence, candor,

fidelity, and accountability. That is not the end of expertise. It is a more demanding, more transparent, and perhaps more worthy form of it.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

Dr Krumholz has received stock options from Element Science, OpenEvidence, Arboretum, and Identifeye, and payments from Arboretum and F-Prime for advisory roles over the past 3 years; was co-editor of Journal Watch Cardiology of the Massachusetts Medical Society; was a section editor of UpToDate; was a cofounder of and held equity in Hugo Health; is a cofounder of and holds equity in Refactor Health, Carabel, and ENSIGHT-AI; is a cofounder of

medRxiv; and serves on the board of openRxiv (nonpaid, volunteer). Through Yale University, he is associated with research contracts from Johnson & Johnson, Novartis, and Pfizer, and research grants from the National Institutes of Health; through Yale New Haven Hospital, he is associated with contracts from the Centers for Medicare & Medicaid Services.

ADDRESS FOR CORRESPONDENCE: Dr Harlan M. Krumholz, Editor-in-Chief, *JACC*, Yale School of Medicine, 195 Church Street, Fifth Floor, New Haven, Connecticut 06510, USA. E-mail: hkrumholz@acc.org.

REFERENCES

1. Montero A, Montalvo J III, Kearney A, et al. KFF tracking poll on health information and trust: use of AI for health information and advice. KFF; Published March 25, 2026. Accessed August 16, 2026. <https://www.kff.org/public-opinion/kff-tracking-poll-on-health-information-and-trust-use-of-ai-for-health-information-and-advice/>
2. Ariss SM. Asymmetrical knowledge claims in general practice consultations with frequently attending patients: limitations and opportunities for patient participation. *Soc Sci Med*. 2009;69: 908-919. <https://doi.org/10.1016/j.socscimed.2009.06.045>
3. Nori H, Daswani M, Kelly C, et al. Sequential diagnosis with language models. *arXiv*. Published online July 2, 2025. <https://doi.org/10.48550/arXiv.2506.22405>
4. Jabbour S, Fouhey D, Shepard S, et al. Measuring the impact of AI in the diagnosis of hospitalized patients: a randomized clinical vignette survey study. *JAMA*. 2023;330:2275-2284. <https://doi.org/10.1001/jama.2023.22295>
5. Khera R, Simon MA, Ross JS. Automation bias and assistive AI: risk of harm from AI-driven clinical decision support. *JAMA*. 2023;330:2255-2257.
6. Cappella JN, Street RL Jr. Delivering effective messages in the patient-clinician encounter. *JAMA*. 2024;331:792-793. <https://doi.org/10.1001/jama.2024.0371>
7. Sauerbrei A, Kerasidou A, Lucivero F, Hallowell N. The impact of artificial intelligence on the person-centred, doctor-patient relationship: some problems and solutions. *BMC Med Inform Decis Mak*. 2023;23:73. <https://doi.org/10.1186/s12911-023-02162-y>
8. Duong T, Plage S, Woods L, et al. Consumer perspectives on trust in and benefits of artificial intelligence in health care. *JAMA Netw Open*. 2026;9:e2626916. <https://doi.org/10.1001/jama-networkopen.2026.26916>